

Biniyam Lombe

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SUMMARY

Software engineer and ML researcher who ships full-stack products end-to-end and builds ML systems from data pipeline through trained model to production inference service. Yale M.S. in Computer Science; comfortable moving between backend/database engineering, computer vision, and NLP/LLM research.

EDUCATION

Yale University New Haven, CT
M.S. in Computer Science | *GPA: 3.91/4.0* | *Wade Scholar (\$50,000)* *Aug. 2024 – May 2026*
– Coursework: Machine Learning, Deep Learning Theory & Applications, Large Language Models, Computer Vision, Database Systems, Algorithms, Artificial Intelligence

University of Bristol Bristol, UK
B.Eng. Electrical & Electronic Engineering | *First Class Honours* *Sep. 2019 – Jun. 2022*

EXPERIENCE

Machine Learning Research Assistant New Haven, CT
Yale Vision Lab *Jun. 2025 – Jan. 2026*
– Built end-to-end PyTorch training pipelines (ResNet, VGG, U-Net) for real-time surgical instrument tracking, achieving **96% F1-score** on held-out video.
– Designed a versioned data pipeline over **29,400+ video frames** with automated quality validation, making training runs reproducible; tracked experiments in **MLflow** and deployed models as **FastAPI** inference services.

Graduate Teaching Fellow New Haven, CT
Yale School of Engineering & Applied Science *Aug. 2024 – May 2026*
– Taught **300+ students** across Data Structures, Database Systems, Machine Learning-adjacent, and AI for Future Presidents; built automated grading pipelines in Python and C that cut turnaround by **40%**.

SOFTWARE ENGINEERING

Waylo Go | *React, TanStack Start, TypeScript, Supabase, PostgreSQL, Stripe, Cloudflare Workers* Jan. 2026 – Present
– Built and deployed a full-stack logistics marketplace **solo**: server APIs, real-time chat, role-based auth, and server-side authorization, with no team to hand off to.
– Enforced **Row Level Security** across all tenant-facing tables; hardened auth with hashed/expiring tokens, **constant-time comparisons** to close a timing side-channel, and rate limiting.
– Set up typed server functions with Zod validation and migration-based schema management; integrated Stripe billing/webhooks and Twilio OTP verification.

ApplyMate: LLM-Powered Matching Engine | *Next.js, TypeScript, PostgreSQL, Gemini API* 2025 – Present
– Shipped a **backend REST API integrating an LLM** (Gemini): an extraction pipeline converting unstructured text into structured, rankable JSONB data, combined with rule-based scoring.
– Implemented **graceful model fallback** so the system degrades safely when an inference call fails; built role-based middleware, SSR session handling, and Stripe subscription billing.

PathReview | *Open-source AI portfolio review platform — Python, FastAPI, SQLAlchemy*
– Restored full green status for review-service unit tests (19/19) by fixing async **SQLAlchemy** mock contracts that made `.scalars().first().all()` fail under pytest.
– Submitted an **upstream open-source pull request** with root-cause analysis, a minimal test-only diff, and reviewer-ready reproduction steps.

PostgreSQL Library Management System | *PostgreSQL, Python, PL/pgSQL, Docker* Oct. – Nov. 2024
– Designed a normalized **3NF schema** across 10+ tables; achieved a **4.6× query speedup** through index tuning; implemented PL/pgSQL triggers for automated business-rule enforcement.

MACHINE LEARNING & AI RESEARCH

Scaling Laws Break Down for Low-Resource Languages | Researcher, Yale Jan. – May 2026
– Built a **decoder-only transformer from scratch in PyTorch** (200K–103M params, no pretrained backbone) and ran a **5×5×3 factorial study** (75 runs per language) across a **SLURM-managed GPU cluster**.
– Enforced a leakage-free evaluation protocol (token-disjoint splits, measured token accounting); found standard compute-optimal scaling assumptions break down in this regime and reported the result transparently.

Statistical Robustness & Fairness Benchmarking | *Python, DeepFace, TensorFlow, Fairlearn* Sep. – Dec. 2025

- Built a reusable evaluation framework benchmarking **7 pretrained models** on 938 verification pairs under controlled degradation, quantifying effect sizes (AUC drops of 0.07–0.09) and finding fairness gaps can **invert direction** under real-world conditions.

Semantic Segmentation with U-Net, CVAT & SAM2 | *PyTorch, CVAT, SAM2* Nov. – Dec. 2025

- Trained a **U-Net** from scratch in PyTorch, reaching **81.1% pixel accuracy and 60.5% mIoU**; built the annotation pipeline in CVAT using **Segment Anything (SAM2)** for mask generation.

Real-Time ASL Recognition | *YOLOv8, Python, OpenCV, CUDA (via PyTorch)* CPSC4800, Yale

- Built a real-time, two-stage **YOLOv8** pipeline (hand detection + letter classification) with temporal smoothing over prediction history and live FPS tracking.

RESEARCH COMMUNITY

Peer Reviewer, NeurIPS 2026 (Conference & Evaluations and Datasets Track), ICML 2026 Workshop, ACL 2026 Workshop.

Judge, Yale Undergraduate Research Conference, evaluated technical research from 400+ researchers across 110+ universities.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C, Bash

ML & Deep Learning: PyTorch, TensorFlow, scikit-learn, transformer architectures, LLM fine-tuning, CNNs (ResNet, VGG, U-Net, YOLOv8), semantic segmentation, tokenization (BPE)

Computer Vision: DeepFace, U-Net, CVAT, SAM2, YOLOv8, OpenCV, image classification, demographic/fairness evaluation

MLOps & Infrastructure: MLflow, FastAPI, Flask, Docker, SLURM/GPU cluster training, versioned data pipelines, experiment tracking

Frontend & Backend: React, Next.js, TanStack Start, Tailwind CSS, FastAPI, Flask, REST APIs, Zod validation

Databases: PostgreSQL, Supabase, MySQL, schema design, query optimization, Row Level Security, PL/pgSQL, ETL pipeline design

Auth, Payments & Security: Supabase Auth, Stripe, Twilio, OWASP, authentication hardening, timing-attack mitigation

Tools: Git, Docker, Cloudflare Workers, GCP, Vercel

Certifications: Google Cybersecurity Certificate